BAURods
Global Sensor Characterization

Technical report on the integrated 3 × 3 visuotactile sensing skin

Prepared 19 October 2026 | Manual acquisition: 8 August 2026

Reconciled edition | 19 October 2026. Acquisition: 8 August 2026, session S3. This edition supersedes the 6 October characterization narrative and aligns with the latest complete model report [S1] and data-quality report, including its 18 October statistical supplement [S2].

BAURods demonstrates useful model-assisted contact discrimination and nine-region localization across one integrated 3 × 3 visuotactile skin. The latest retrospectively ranked models achieved **99.24% contact accuracy**, **84.88% localization accuracy**, and **0.4972 N force MAE** under grouped evaluation. The physical characterization establishes a measurable global optical response across manual loading, while the quality supplement quantifies recording-level variability and its uncertainty. These are complementary findings from the same archive, not independent replications.

The strongest defensible conclusion is that the integrated optical sensing system contains useful contact and spatial information and supports moderate force estimation within the evaluated 0.05-3 N interval. Its global optical response alone is not a precise force calibration. Controlled fixed-force repeatability, intrinsic response time, and independent-day reproducibility remain unestablished. No application-specific acceptance limit has been supplied, so this report makes no pass/fail claim.

1. Evidence scope and cross-report reconciliation

Table 1

Item	Reconciled interpretation	Reason for different reported values
Latest model headlines	Best evaluated families ranked retrospectively by classification accuracy or force MAE	Earlier nested-selector results evaluate a different selection procedure
72 press recordings; 81 contact runs; 5,357 pulses	Recording count, connected threshold runs, and finer algorithm-inferred loading pulses	None is a verified count of individual physical finger presses
Model sample	54 press recordings; six TEST groups; 7,154 eligible frames	Broader characterization includes 18 replay-designated recordings excluded from this model protocol
Contact model sample	8,506 frames in 65 recordings, including 11 no-contact recordings	Frame metrics differ from event/window or archive-wide gate metrics
Baselines	Four baseline captures in the full archive; three in the model subset	Different subsets, not conflicting provenance
Optical response	Global magnified positive-clipped peak; target-ROI recording mean; original-video signed correction	Different processing and aggregation produce different slopes and variances
R^2	0.2229 global optical fit; 0.3113 held-out force prediction; 0.7617 adjusted recording model	Different dependent variables, covariates, units of analysis, and evaluation methods
Gate metrics	0.127% archive contact-frame accuracy; 0.597% pulse detection recall	Different denominators; neither is the learned detector's 99.24% accuracy

The user confirmed separate repeated finger presses. This supports the experimental interpretation but does not validate the segmentation count. Pulses can occur without force returning below 0.05 N. Consequently, 81 contiguous threshold-connected contact runs can contain many local loading peaks. Use “algorithm-inferred loading pulses” for the 5,357 retained segments throughout thesis tables and discussion.

2. Global response definition and acquisition

The archive contains 83 recordings: 72 press recordings and 11 dedicated no-contact recordings, with 29,318 synchronized frames and 42,568 raw force observations. All belong to one session. Nine taxels form one sensing skin; taxel-specific results are secondary spatial diagnostics, not nine independently validated sensors.

The global optical statistic S is the maximum over each accepted pulse window of the maximum across nine ROI means of positive per-pixel baseline-corrected HSV V. It is measured on archived color-magnified images (amplification 30; passband 1.1-1.2 Hz). V denotes image digital units, not volts or calibrated radiance. Force F is the raw reference peak associated with that window. The optical and force maxima need not occur at the same instant, so their relationship is a pulse-amplitude association rather than an instantaneous transfer law.

Force-only segmentation used 0.5 N peak prominence, minimum 0.15 s peak separation, and at least three camera frames per accepted window. It retained 5,357 pulses; 2,000 candidates were rejected, mostly because fewer than three frames were available. Observations remain clustered within recordings. The physical press count is unverified.

3. Global force-response characteristic

Table 2

Characteristic	Estimate	Scope
Observed peak-force span	1.1512-23.3004 N	Retained pulse maxima; not rated capacity
Optical-response span	0.7071-8.5698 V units	Global magnified positive-clipped pulse response
Linear response	S = 0.90855 + 0.12106 F	Pooled descriptive fit
Apparent sensitivity	0.12106 V units/N	Protocol-dependent slope
Slope 95% interval	0.08885-0.14901 V units/N	Recording-cluster bootstrap
In-sample fit	R^2 0.2229; RMSE 0.7045 V units	Predicting optical response from force
Recording-held-out fit	R^2 0.1934; RMSE 0.7178 V units	Model-selection cross-validation, not untouched final validation
Residual dispersion	SD 0.7046; IQR 0.8394 V units	Manual-loading pulse-level variability

The positive pooled trend supports an observable system response to loading. Its scatter limits a single global amplitude-to-force calibration. The 23.3004 N raw-force peak and the quality report's 22.699 N synchronized-force maximum use different sampling streams; the camera-aligned stream can miss raw-force extrema. The model's 0.05-3 N eligibility interval is narrower than both, and its reported prediction error cannot be extended to the full peak-force span.

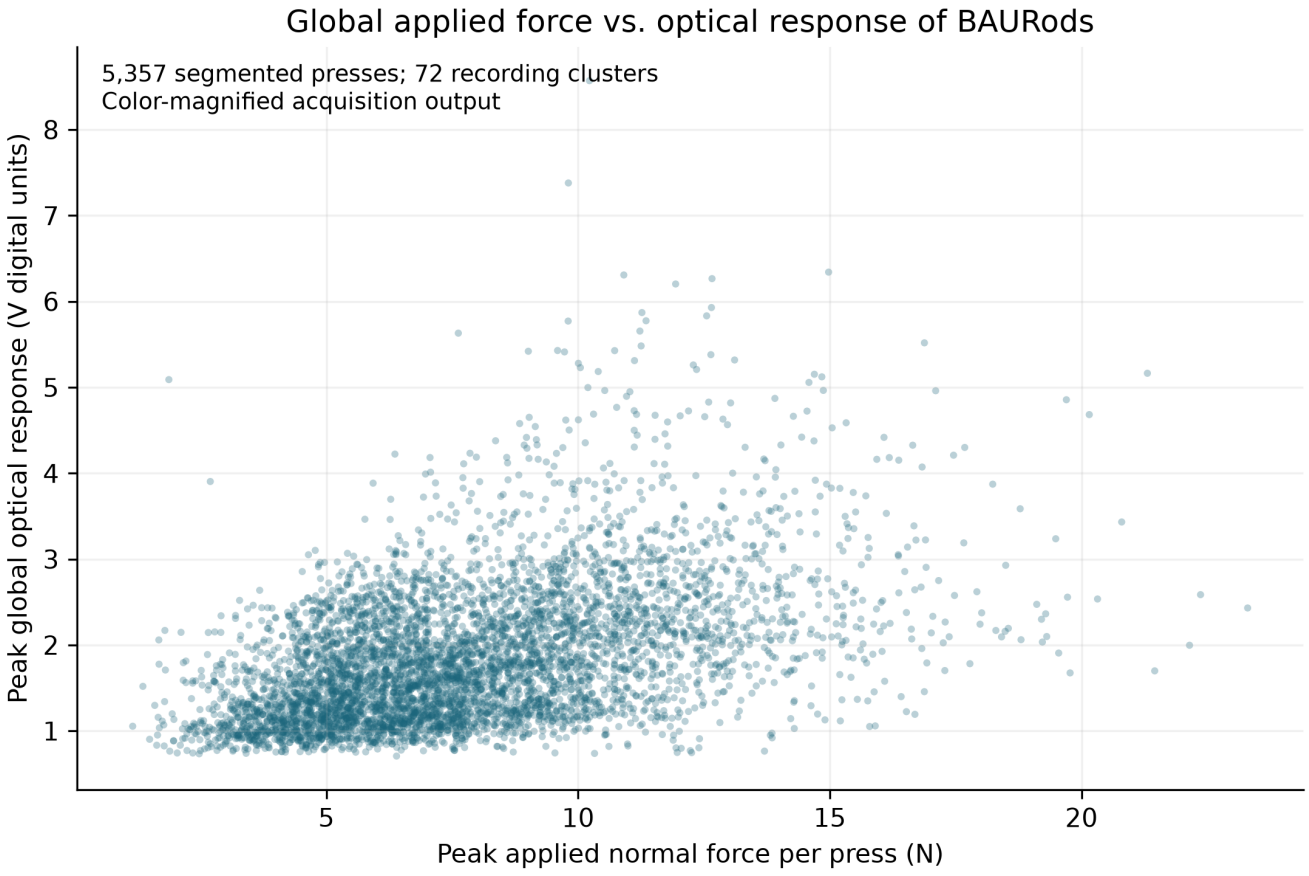


Figure 1. Global applied peak force and optical response of the complete BAURods skin. Each point represents one retained algorithm-inferred loading pulse; observations remain clustered in recordings.

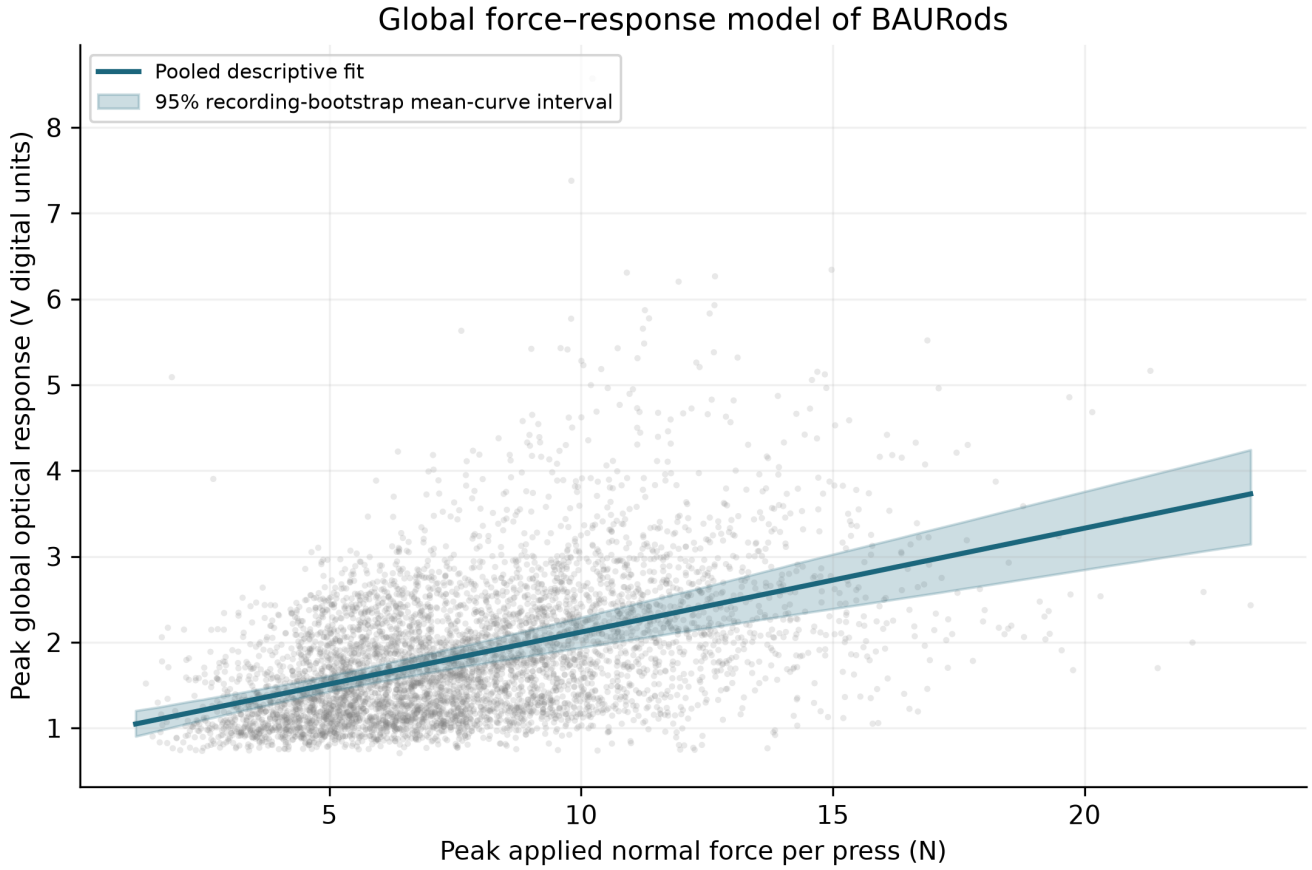


Figure 2. Global fitted force-response characteristic. The model describes archived optical response in V digital units; its slope is not an intrinsic material sensitivity.

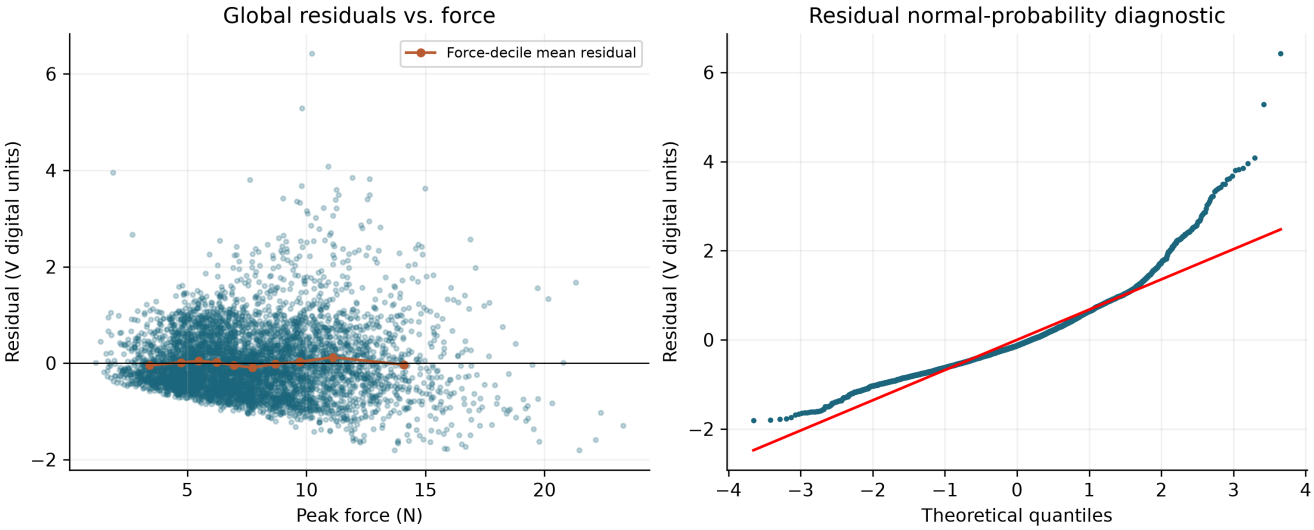


Figure 3. Residual diagnostics for the global force-response model.

4. Response dispersion under manual loading

Pulse-level residual SD of 0.7046 V units describes deviations from the global line under varying force, position, baseline context, and manual loading. It is a conditional dispersion proxy, not fixed-force sensor repeatability. The response distribution and force-conditioned spread should be interpreted alongside those uncontrolled factors. Selecting only visually favorable pulses would change the estimand and is not used here.

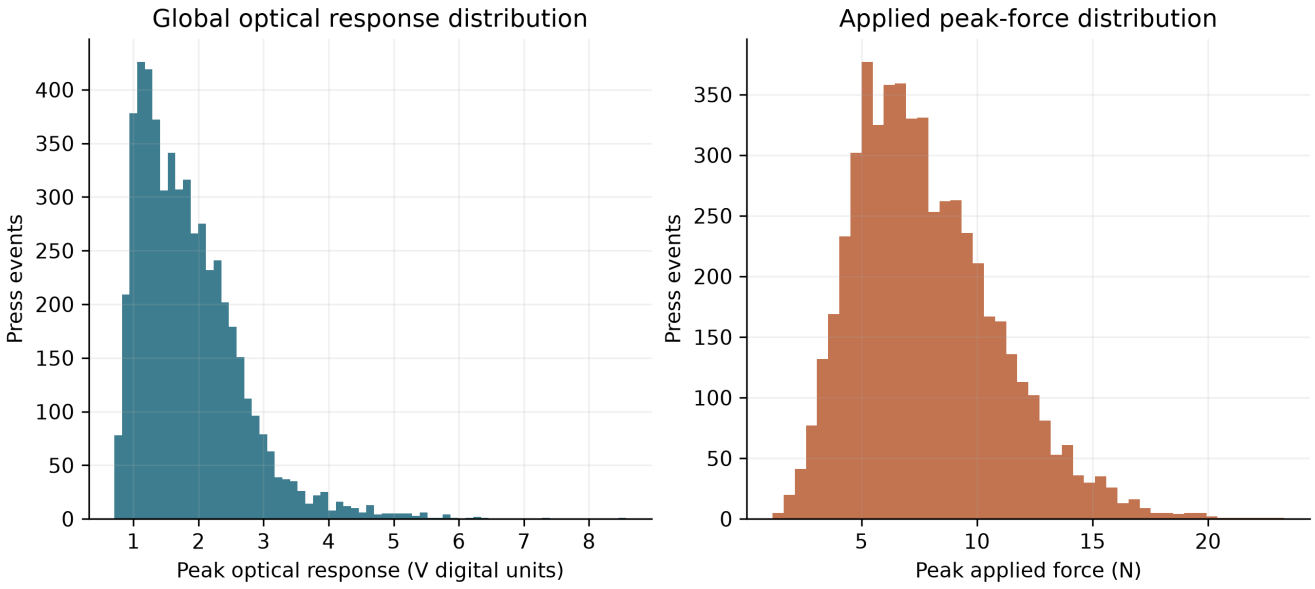


Figure 4. Distribution of global optical response across retained loading pulses.

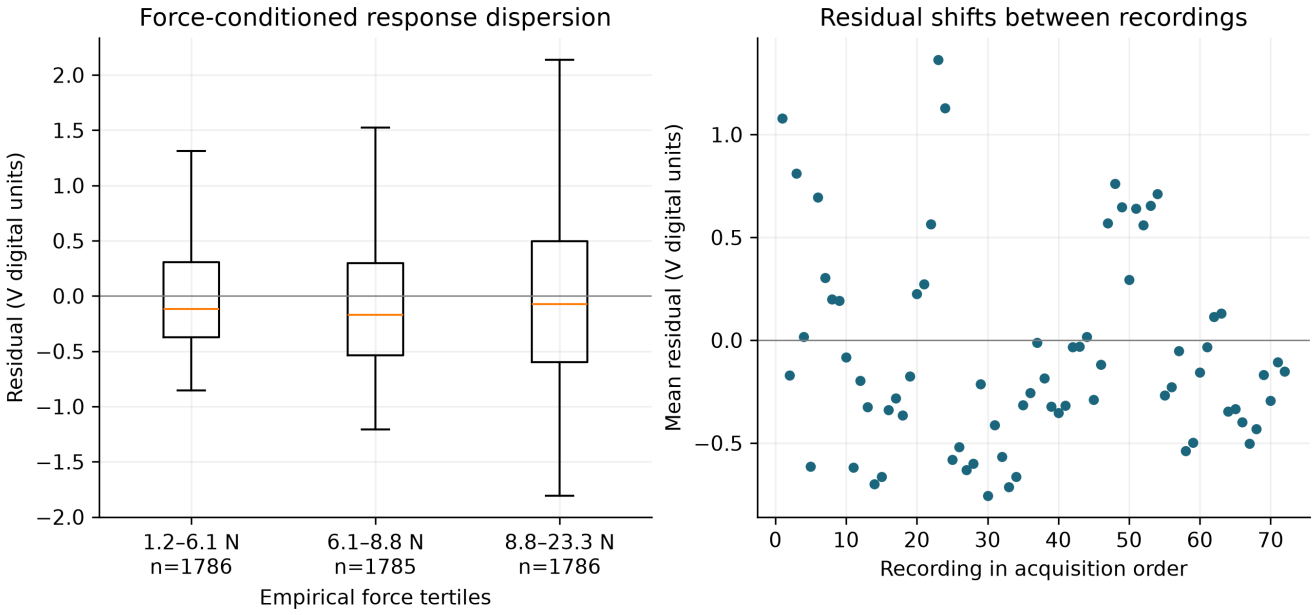


Figure 5. Force-conditioned response dispersion under manual pressing. This is a repeatability proxy, not controlled fixed-force metrological repeatability.

5. Updated recording-level repeatability evidence

The quality supplement [S2] uses 72 recording means: nine taxels across eight TEST groups. Each response is the labelled target ROI's mean positive pixel delta-V, averaged over valid force-derived contact frames (force at least 0.05 N). Each recording has one observation and equal weight. Between-recording optical CV ranges from **9.45% to 28.66%**, with eight observations per taxel. This quantifies within-session variation under manual loading.

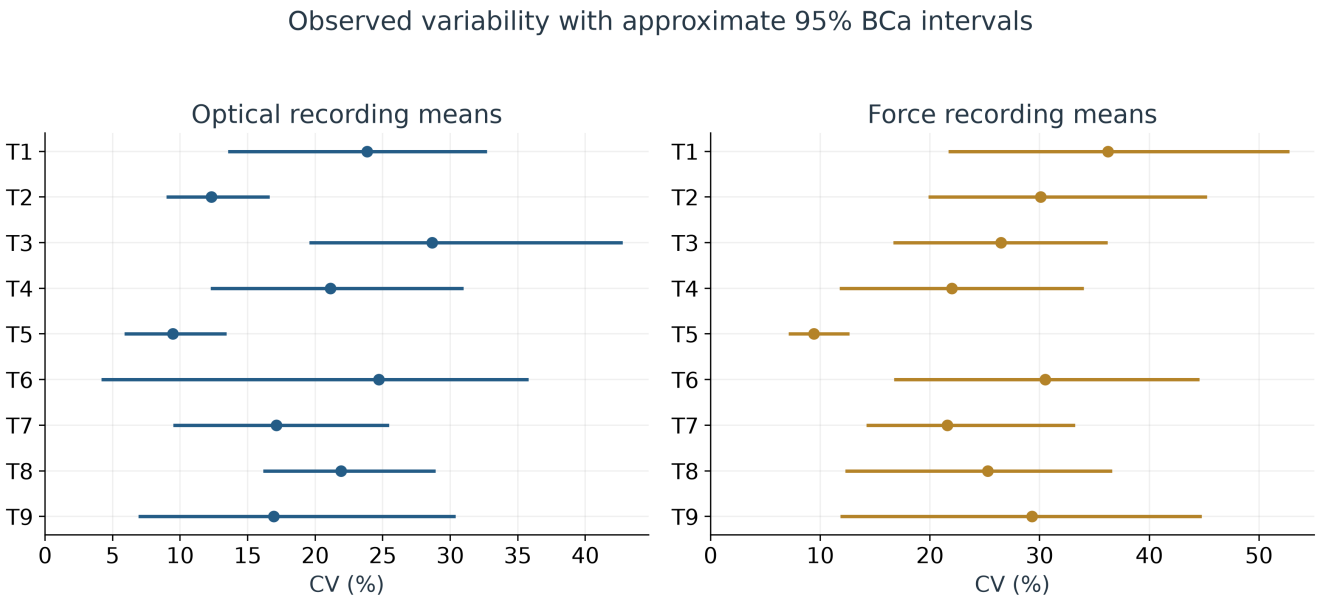
The adjusted model includes an intercept, taxel effects, TEST-group effects, and mean force centered at 3.5 N. It has 17 parameters and 55 residual degrees of freedom. The common overlap of recording-mean force across taxels is approximately 2.976-3.812 N; 3.5 N is an adjustment reference, not a newly performed fixed-force test.

Table 3

Recording-level statistic	Estimate	Uncertainty or interpretation
Pooled within-taxel SD	0.2671 delta-V	95% BCa interval 0.2126-0.3464
Taxel/TEST-adjusted SD before force term	0.2063 delta-V	Residual variation after blocking
Force-adjusted residual SD	0.1946 delta-V	95% BCa interval 0.1597-0.2352
Common force association	0.0811 delta-V/N	Primary TEST-group jackknife 95% interval -0.0087-0.1708
Full model R^2	0.7617	In-sample variance explained jointly by all terms
Partial R^2 for force	0.1261	Incremental explanation relative to blocked model

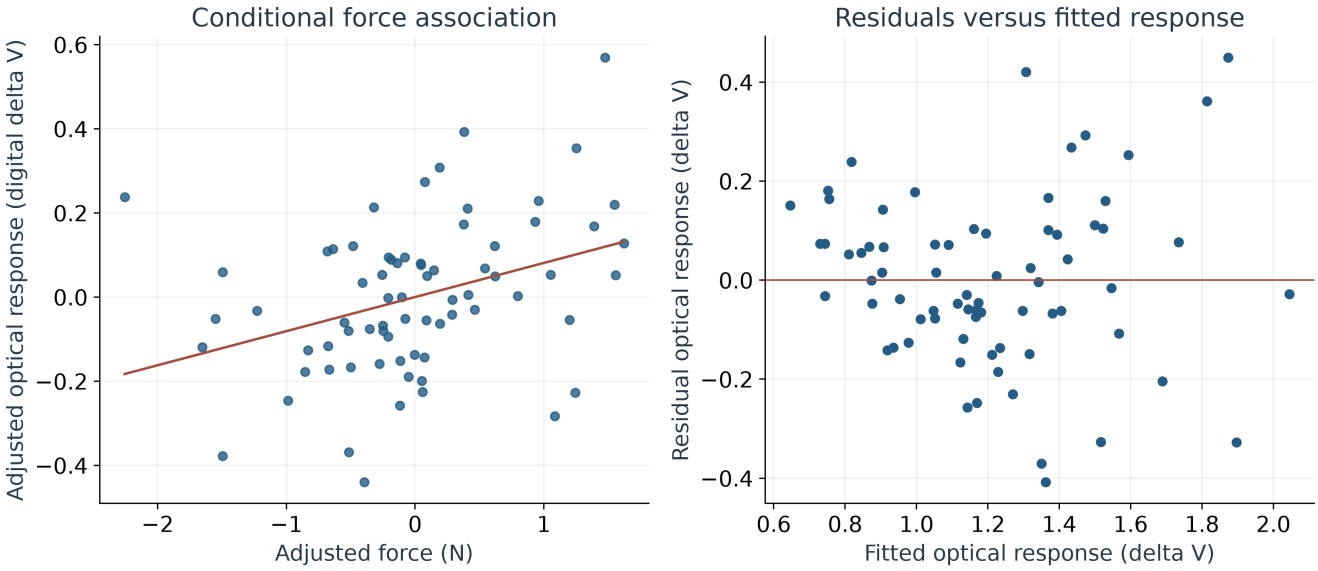
The force term reduces residual SD by about 5.7%, from 0.2063 to 0.1946. The full R^2 of 0.7617 is not force-prediction accuracy and does not mean force alone explains 76.17% of response variation. The primary delete-one-TEST-group jackknife uses eight clusters and a t interval with seven degrees of freedom; it includes zero. The whole-group BCa slope interval (0.0244-0.1619) excludes zero, while the taxel-jackknife interval also includes zero. Thus inference about a common force association depends on the uncertainty method. The source uses 50,000 paired whole-group bootstrap resamples, retaining all nine taxels in each sampled group rather than resampling individual frames.

The global pulse slope's positive interval and the adjusted recording slope's zero-crossing interval do not contradict each other: the response statistic, aggregation, covariates, weighting, and uncertainty procedures differ. Neither establishes an intrinsic material sensitivity. No acceptance threshold or sufficient-repeatability verdict is inferred from the observed CVs.



Eight recordings per taxel; 50,000 whole-group resamples. No acceptance threshold.

Recording-level variability and uncertainty, reproduced from the quality supplement



Force-adjusted recording diagnostics, reproduced from the quality supplement

6. Baseline stability and noise

Across the 11 unloaded recordings, the global maximum-ROI statistic has a recording-equal mean of 0.8980 V units, RMS within-recording SD of 0.04372 V units, and temporal CV proxy of 4.87%. Between-recording SD is 0.03316 V units. The median pulse-excess-to-baseline-noise ratio is 18.41; it is a processing-dependent SNR proxy, not a physical signal-to-noise specification or detection limit.

The quality report instead evaluates 99 individual recording/ROI traces. Raw mean-V CV spans 0.312-0.725%; positive-corrected ROI CV spans 5.50-9.14%, with means 0.502-0.915 delta-V. These are different statistics from the global maximum-ROI trace. Positive clipping gives a nonzero unloaded response. The observed short recordings support short-term baseline stability only, not stability across independent days or environmental conditions.

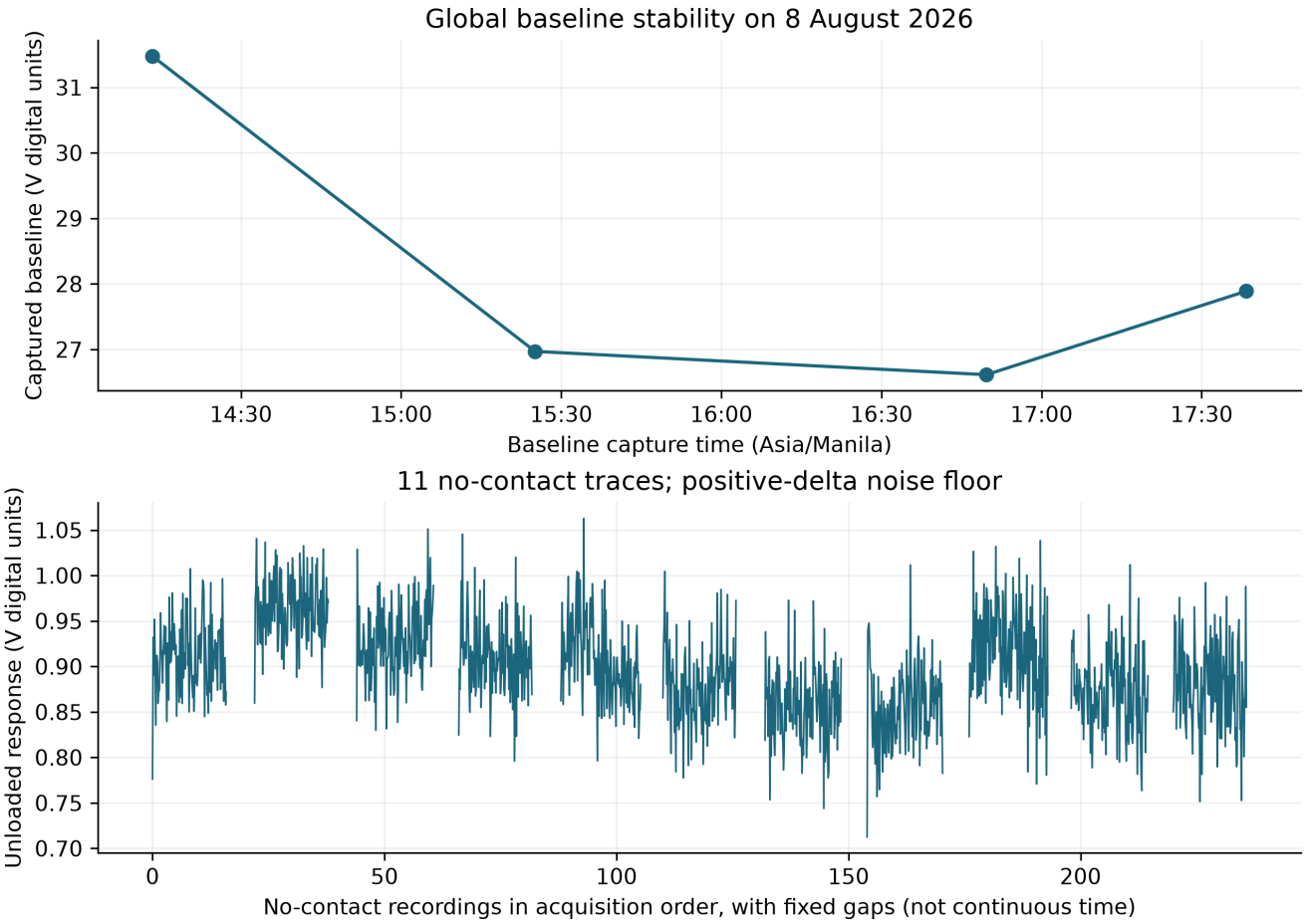


Figure 6. Recorded baseline stability and unloaded optical traces.

7. Temporal behavior and unresolved physical metrics

Median camera spacing is approximately 0.1301 s (about 7.69 Hz), compared with 0.09188 s for raw force. A retained pulse has a median of only four camera frames, and only one accepted event has unloaded boundaries on both sides. Filtering and sparse sampling prevent defensible intrinsic response-time, recovery-time, hysteresis, or bandwidth estimates. The sampled traces are useful descriptions of observed behavior, not step-response tests.

No saturation limit, minimum resolvable force, calibrated detection limit, or independent-day reproducibility is established. The previously reported 41.10 ms timing refers to an offline audit of the original inference bundle. It is neither sensor response time nor a measured latency of the latest winning-model combination.

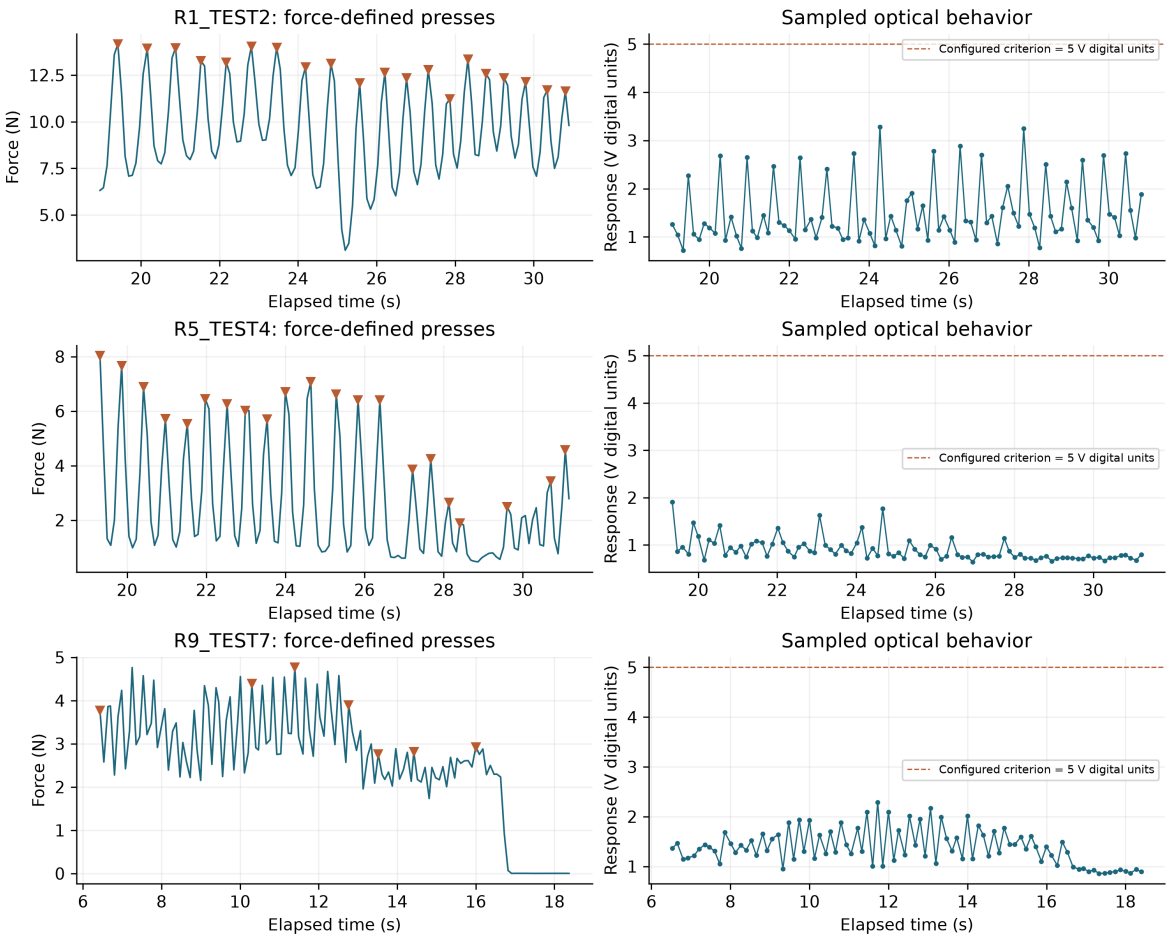


Figure 7. Sampled force and optical behavior during repeated manual presses. Temporal filtering and sparse samples prevent inference of intrinsic response or recovery time.

8. Archived gate and spatial information

The saved 5-V acquisition gate detects 32 of 5,357 retained pulses, giving **0.597% pulse recall**, with 5,325 missed pulses. All 11 unloaded recording windows are negative. Specificity is 100% on those 11 windows, but this small control set does not establish a population false-positive rate. Combining pulse and unloaded-window units yields 0.801% diagnostic accuracy; it is not comparable to a frame-weighted model score.

The quality report's archived localization accuracy/coverage of **0.127%** is evaluated on contact frames. The pulse-level gate and localization calculation uses pulse maxima, so its denominator and sampling differ. These low archived-gate results are retained transparently and do not describe the subsequently trained detector.

When contact is assumed known and the global peak is localized by ungated ROI argmax, pulse-level localization is 94.31% (macro F1 0.9444). This favorable spatial result shows useful location information, but it is conditional on known contact and uses different features and units from the learned 84.88% frame-level localization result. It must not be advertised as end-to-end contact-and-location accuracy.

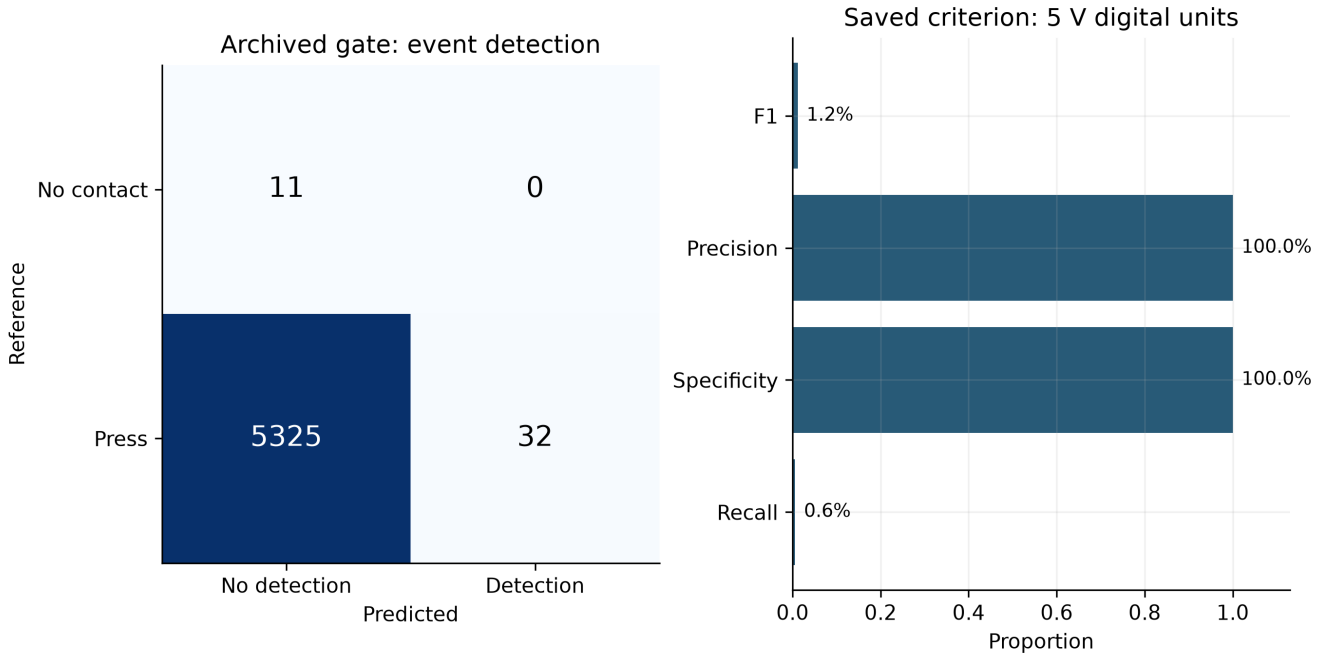


Figure 8. Archived 5-V acquisition gate: 5,357 inferred loading pulses and 11 no-contact recording windows. These event/window metrics are separate from the learned frame-level detector.

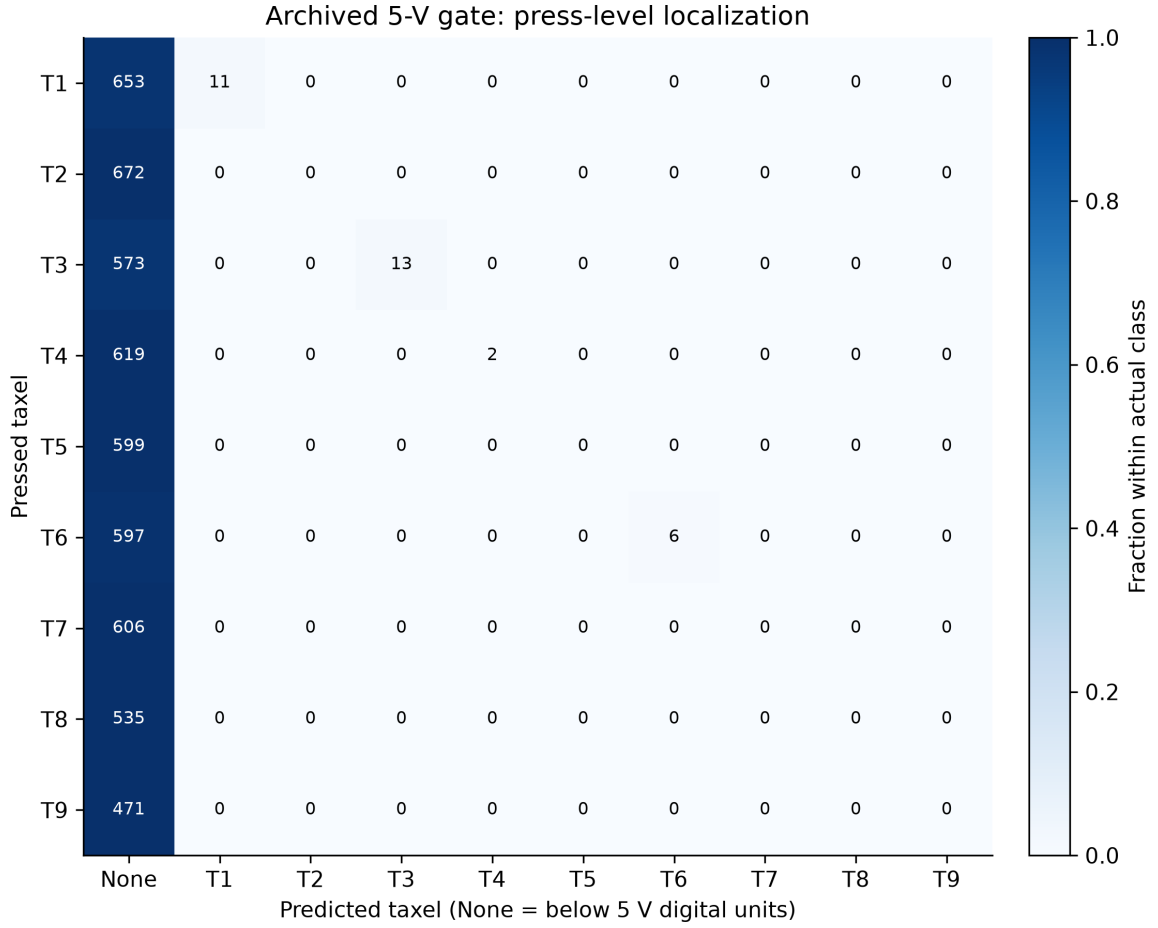


Figure 9. Archived-gate pulse-level localization. No-detection outcomes remain in the denominator; this matrix does not describe the later learned localization model.

9. Updated model-assisted sensing performance

The latest complete model report [S1] selects the highest-accuracy classification families and the lowest-MAE force family among evaluated models. These winners were ranked retrospectively using outer evaluation results. Their predictions come from six complete held-out TEST groups with equal total weight per recording. Family ranking on those results does not create an untouched prospective test set.

Table 4

Task and selected model	Principal result	Supporting metrics
Contact: tuned RBF SVC	Accuracy 99.24%	Balanced accuracy 99.54%; macro F1 0.9867
Nine-region localization: tuned logistic regression	Accuracy 84.88%	Balanced accuracy 84.88%; macro F1 0.8500
Force: tuned ensemble	MAE 0.4972 N	RMSE 0.6299 N; R^2 0.3113; bias +0.0827 N

Contact recall is 99.09%, with 100% observed precision and specificity in this evaluation. Contact uses 8,506 frames from 65 recordings: 7,154 eligible press frames from 54 recordings and 1,352 no-contact frames from 11 recordings. Force and localization use the 7,154 press frames, restricted to 0.05-3 N. All use original, unannotated video features. Correlated frames are not independent experimental repetitions. The learned pipelines were evaluated offline; these scores do not imply deployment into the archived acquisition GUI.

Table 5

Evaluation policy	Contact accuracy	Localization accuracy	Force mae
Latest retrospectively ranked families	99.24%	84.88%	0.4972 N
Previously reported nested selector	97.51%	84.40%	0.4991 N

The earlier nested selector's contact balanced accuracy was 98.50%, and its joint correct-contact-and-region score was 83.61%. Those remain historical results of that procedure. They are not scores of the current winning-model combination. No new joint score or winner-combination timing is inferred in this revision. Localization macro F1 of 0.8500 belongs to the accuracy-winning logistic model; the ensemble's slightly higher macro F1 must not be combined with logistic accuracy as though they described one model.

10. Original-video baseline correction and normalization

The model report's new optical analysis uses original-video ROI mean V, the saved baseline mean V0, signed correction V - V0, and normalized correction 100 × (V - V0)/V0. It includes the same 7,154 eligible frames and 54 recordings, with equal recording weights within each taxel. This signed difference of means is distinct from the archived mean of positive per-pixel differences on magnified images.

The following descriptive R^2 values are squared weighted correlations of each optical signal with force, not held-out force-prediction R^2. Baseline correction improves pooled association for eight of nine taxels; T1 decreases. Normalization adds only modest changes and does not itself establish improved trained-model accuracy.

Table 6

Taxel	Raw v r^2	Signed corrected r^2	Normalized r^2
T1	0.0322	0.0036	0.0052
T2	0.0000	0.0775	0.0775
T3	0.0994	0.2963	0.3026
T4	0.0288	0.0660	0.0654
T5	0.0003	0.0768	0.0735
T6	0.0881	0.1059	0.1122
T7	0.0002	0.0379	0.0389
T8	0.0121	0.1250	0.1262
T9	0.0187	0.1702	0.1755

Subtracting a fixed baseline and dividing by a positive fixed baseline preserves within-recording correlation. Changes in pooled correlations therefore concern between-recording offsets and scaling. The three saved baseline captures used by this subset are consistent with four captures in the broader archive.

11. Interpretation for thesis use

The results support BAURods as an integrated optical sensing skin with useful spatial discrimination and strong retrospective learned contact classification under the evaluated conditions. Force estimation is supported over 0.05-3 N with about 0.50 N MAE, while moderate force R^2 and broad global-response scatter limit claims of precision. Baseline correction helps harmonize original-video optical measurements across most locations without resolving all location dependence.

The recording-level supplement strengthens the characterization by adding uncertainty-aware variability estimates. It does not convert manual loading into a controlled repeatability test. One session, shared baselines, unknown physical-press count, processing-dependent brightness, and family selection using evaluation results constrain generalization. These limitations qualify the demonstrated capabilities without negating the useful contact and spatial information.

12. Characterization summary

Use the current winning-model metrics with their retrospective ranking and 0.05-3 N labels; retain the nested-selector scores only when discussing that evaluation procedure. Use 5,357 algorithm-inferred pulses, 81 threshold-connected contact runs, and 72 press recordings as separate quantities. Use 0.1946 delta-V as the adjusted recording residual SD and 0.7046 V units as the global pulse residual SD, always naming the response and unit of analysis.

For future independent validation, specify application tolerances before testing, control loading and unloading, annotate physical presses, and collect separate sessions. This edition reconciles existing evidence; it does not retrain models, add experimental observations, or establish new acceptance criteria.

Appendix A. Thesis-ready Results and Discussion

The integrated BAURods skin exhibited a positive pooled relationship between force-pulse amplitude and maximum optical response, with an apparent slope of 0.12106 V digital units/N and R^2 of 0.2229. This analysis included 5,357 algorithm-inferred loading pulses across 72 recordings; the physical press count was not independently annotated. The relationship characterizes the archived magnified optical pipeline under manual loading rather than an intrinsic material calibration.

Across eight recording-level repetitions per taxel, optical mean-response CVs ranged from 9.45% to 28.66%. A model adjusting for taxel, TEST group and mean force yielded a residual SD of 0.1946 delta-V, with an approximate 95% BCa interval of 0.1597-0.2352. Its common force coefficient was 0.0811 delta-V/N, with a primary group-jackknife interval of -0.0087 to 0.1708. These findings quantify within-session recording variability; no fixed-force repeatability acceptance criterion or pass/fail judgment was applied.

Using original-video features and equal recording weights across six held-out TEST groups, the retrospectively ranked winning models achieved 99.24% contact accuracy, 84.88% nine-region localization accuracy and 0.4972 N force MAE. Force and localization were evaluated on 7,154 eligible frames from 54 recordings within 0.05-3 N; contact added 1,352 frames from 11 no-contact recordings. These results differ from the previously reported nested-selector scores because they evaluate a different model-selection policy. They are retrospective evidence, not prospective independent-session validation.

Together, these findings support useful contact and spatial information in the integrated sensing skin and moderate force-estimation capability over the evaluated interval. Differences among pulse-level, recording-level and frame-level metrics arise from their response definitions, sample restrictions and statistical estimands. Controlled loading, independently annotated presses and additional acquisition sessions are needed to establish intrinsic dynamic performance and metrological repeatability.

Appendix B. Source register and validation

[S1] Manual Calibration Complete Thesis Technical Report, latest saved edition 19 October 2026 (folder dated 18 October), 59 pages. Includes retrospective winning-model comparisons and original-video baseline-corrected/normalized V analyses. Numerical source: reports/BEST_RESULTS/BEST_METRICS.json and best_models_held_out_predictions.parquet.

[S2] BAURods Manual Calibration Technical Report, 39 pages, including Part VIII statistical supplement dated 18 October 2026. Numerical source: baurods_quality/outputs/repeatability_20261018/summary.json and tables/analysis_recordings.csv.

[S3] Manual Calibration (2).zip. SHA-256: e039dcd49f3390d34dc979490dcf46a6ecc43d9e6a2aec08e4952acc14b5006a. Acquisition: session S3, 8 August 2026. Original global characterization outputs and figures are retained; this revision changes reporting and integrates updated evidence.

[S4] Original-video signal comparison: reports/manual_normalized_v_force_2026-10-19/signal_comparison_by_taxel.csv. Three reused baseline captures; signed correction and normalization are distinct from positive-clipped magnified response.

Revision verification: independently recomputed classification accuracy, balanced accuracy and macro F1, plus force MAE, RMSE, bias and R^2 from saved held-out predictions. All ten metrics agree with BEST_METRICS.json within 1e-10. Independently refitted the 72-recording, rank-17 adjusted model; slope and residual SD agree within 1e-10. Source hashes and checks are saved in reconciliation_validation_20261019.json. These are consistency checks, not new experimental validation.

Version control: the previous PDF, narrative and PDF builder are preserved in outputs/global_characterization/history/20261006. The older statistics workbook and ZIP package have not been regenerated and should not be used as the revised headline-model source. Neither companion report was modified.

Figures 1-9 reproduce the original physical analysis. References to presses in legacy plot labels mean algorithm-inferred pulse segments, not verified physical-press counts. The two recording-level figures reproduce the quality supplement and retain its taxel-level diagnostic detail. All main conclusions concern the integrated skin.

Units and abbreviations: V, HSV brightness in image digital units (not volts); N, newtons; ROI, region of interest; SD, standard deviation; CI, confidence interval; MAE, mean absolute error; RMSE, root mean squared error; BCa, bias-corrected and accelerated bootstrap.